

## Practical Assignment Sheet (Python-SQLite-Unit 3)

1. Create "Calculator.py" module and write following functions:
  - 1.1. Addition of two numbers
  - 1.2. Subtraction of two numbers
  - 1.3. Multiplication of two numbers
  - 1.4. Division of two numbers
  - 1.5. Greatest number from two numbers
  - 1.6. Smallest number from two numbers
  - 1.7. Percentage of given number
    - 1.7.1. Example: no=50000 and percent=10% then answer=50000\*10%=5000)
2. Create "calculation.py" file. Take input of required numbers and call each function of "Calculator.py" module and display appropriate message.
3. Perform following tasks:
  - 3.1. Create "Sales.db" database.
  - 3.2. Create following table and insert records.
  - 3.3. Salesman (sid, name, city, commission) (primary key = SID)

| Sid  | Name         | City      | Commission |
|------|--------------|-----------|------------|
| 5001 | Mohan Patel  | Anand     | 0.15       |
| 5002 | Nail Shah    | Surat     | 0.13       |
| 5005 | Preet Vyas   | Ahmedabad | 0.11       |
| 5006 | Jeevan Mehta | Navsari   | 0.14       |
| 5003 | Paul Adam    | Nadiad    | 0.12       |
| 5007 | Ramesh Patel | Surat     | 0.13       |

- 3.4. Display 1<sup>st</sup> and 2<sup>nd</sup> record using fetchone() method.
- 3.5. Display data using fetchall() method.
- 3.6. Display 4<sup>th</sup> and 5<sup>th</sup> records using fetchall() method.
- 3.7. Display data without using fetchone() and fetchall() methods.
- 3.8. Display the records of salesmen who are leaving in Surat city.
- 3.9. Display records of salesmen whose name consists letter "M".
- 3.10. Display records of salesmen whose commission is in range of 0.11 to 0.13.
- 3.11. Display records of salesmen of Anand and Navsari city.
- 3.12. Take salesman's SID from user and display his detail.
- 3.13. Modify the commission of 5006 to 0.15
- 3.14. Modify the City to Bharuch whose commission is 0.13.
- 3.15. Delete the records of 5007.
- 3.16. Delete the records of salesman of Bharuch city.
- 3.17. Insert three records into salesman table as per user's input.

## 4. Perform following tasks:

- 4.1. Create following table in same database.
- 4.2. Write trigger which check whether purchase amount is more than 1000 or not.
- 4.3. Insert records.
- 4.4. Order (oid, pur\_amt, ord\_date, sid)(oid=Primary key &sid=Foreign key)

| Oid | Pur_amt | Ord_date   | sid  |
|-----|---------|------------|------|
| 1   | 12000   | 2020-12-05 | 5001 |
| 2   | 42000   | 2020-12-20 | 5002 |
| 3   | 2000    | 2021-02-02 | 5005 |
| 4   | 14000   | 2021-03-23 | 5005 |
| 5   | 23000   | 2021-04-15 | 5003 |
| 6   | 33000   | 2021-05-20 | 5001 |
| 7   | 32000   | 2021-06-22 | 5003 |
| 8   | 23500   | 2021-07-01 | 5003 |
| 9   | 43000   | 2021-07-05 | 5001 |
| 10  | 12000   | 2021-07-15 | 5002 |

- 4.5. Display the order detail by sorting data as per purchase amount.
- 4.6. Display top five order detail.
- 4.7. Display the order information of the month December.
- 4.8. Display order's detail whose purchase amount is in between 10000 to 20000.
- 4.9. Display Order detail with salesman name and city.
- 4.10. Display salesman ID wise total purchase.
- 4.11. Display orders' detail with Class. Class is declared as per following criteria:  
SID=5001 or 5002 -> Class='A'  
SID=5003 or 5005 -> Class='B'  
Otherwise -> Class='C'
- 4.12. Display orders' detail with salesman's commission.
- 4.13. Display all salesman detail who never placed any order.
- 4.14. Display orders' detail as per user's inserted date.
- 4.15. Take input of two dates from user and display order details between these two dates.

## 5. Perform following tasks:

- 5.1. Create following table in same database.

5.2. Write trigger which check whether grade value is in between 100 to 500 or not.

5.3. Insert records.

5.4. Customer(cid,cust\_name,grade) (cid=primary key)

| cid | Name            | grade |
|-----|-----------------|-------|
| C1  | Mohit Patel     | 100   |
| C2  | Geeta Vyas      | 200   |
| C3  | Jaya Patil      | 100   |
| C4  | Vishal Gohel    | 300   |
| C5  | Kartik Goyenka  | 200   |
| C6  | Meera Prajapati | 100   |
| C7  | Veer Vyas       | 300   |
| C8  | Maya Mehta      | 200   |

5.5. Display unique grades of customers.

5.6. Display customers' detail with bonus. Calculate bonus as per following criteria:

Grade=100 -> Bonus = 5000

Grade=200 -> Bonus = 10000

Otherwise -> Bonus = 15000

5.7. Take input from user for grade and display respective data.

5.8. Take input of single letter from user and display the customers' record whose name start with given letter.

5.9. Delete the record of customer whose Customer ID is given by user.

5.10. Update the grade by adding 50 in existing grades of the customer whose ID is given by user.

5.11. Insert 3 records into Customer table as per user's data.

5.12. Display Salesman detail of city specified by user.

5.13. Display order's detail whose purchase amount is greater than the user's inserted purchase amount.

5.14. Display order detail of the salesman whose Salesman ID is given by user.